

Code and data from: Top-down control and species composition mediate the short-term response of experimental food webs to a nutrient pulse perturbation

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Description of the data and file structure

Information about the datasets and the code

The aim of this study is to investigate the response in biovolume of phytoplankton (algae) and zooplankton (rotifers) after a nutrient pulse perturbation. We notably test two types of relationship:

- the hump-shaped relationship between biovolume responses of algae and the top-down control at the pulse perturbation
- the hump-shaped relationship between biovolume responses of rotifers and the top-down control at the pulse perturbation

First, we conducted experiments and then inferred the experimental data via Bayesian inference. Data analysis was performed in R 4.5.0 and RStudio environment using notably the libraries tidyverse, PKNCA, cmdstanr, brms, vegan, gridExtra, scales, readxl, stringr, and deSolve. To run the scripts it is important to keep the structure with the 3 folders otherwise errors can emerge.

The files are organised into three folders: data containing 8 files, scripts containing 10 R scripts and figures containing 8 pdf files. The following sections explain the aim, structure and content of each folder.

Folder data

This folder contains all the files necessary for running the R scripts. NA or empty cell values indicate that data was not available.

- "Chemostat_experimental_timeseries.xlsx" contains the experimental timeseries of species and trophic level biovolumes and of the relative abundance of algal species. The 16 columns are:
 - Category: the category of rotifer treatments "Monocultures" or "Polycultures" (cf. methods)
 - Treatment: the subcategory of treatments: "Bc" for monocultures of *B. calyciflorus*, "Ce" for monocultures of *Cephalodella* sp., "Le" for monocultures of *Lecane* sp; "9d" for rotifer polycultures disturbed 9 days after inoculation, "13d" for rotifer polycultures disturbed 13 days after inoculation.
 - Chemostat: the name of the chemostat

- Date: date of the measurement (DD/MM/YY)
- Time standardised to pulse: day standardised to the day of the pulse perturbation. Negative values correspond to the days before the pulse perturbation, 0 corresponds to the day of the pulse perturbation and positive values correspond to the days after the pulse perturbation.
- Biovolume_total_algae $\mu\text{m}^3/\text{mL}$: total biovolume of the algae species in $\mu\text{m}^3/\text{mL}$
- Biovolume_Cryptomonas $\mu\text{m}^3/\text{mL}$: biovolume of the species *Cryptomonas* sp. in $\mu\text{m}^3/\text{mL}$
- Biovolume_Chlamydomonas $\mu\text{m}^3/\text{mL}$: biovolume of the species *C. reinhardtii* in $\mu\text{m}^3/\text{mL}$
- Biovolume_Monoraphidium_Chlorella $\mu\text{m}^3/\text{mL}$: summed biovolume of the species *M. minutum* and *C. vulgaris* in $\mu\text{m}^3/\text{mL}$
- Biovolume_total_rotifers $\mu\text{m}^3/\text{mL}$: total biovolume of the rotifer species in $\mu\text{m}^3/\text{mL}$
- Biovolume_Brachionus $\mu\text{m}^3/\text{mL}$: biovolume of the species *B. calyciflorus* in $\mu\text{m}^3/\text{mL}$
- Biovolume_Cephalodella $\mu\text{m}^3/\text{mL}$: biovolume of the species *Cephalodella* sp. in $\mu\text{m}^3/\text{mL}$
- Biovolume_Lecane $\mu\text{m}^3/\text{mL}$: biovolume of the species *Lecane* sp. in $\mu\text{m}^3/\text{mL}$
- Relative_abundance_Cryptomonas: proportion of *Cryptomonas* sp. in the algal total abundance
- Relative_abundance_Chlamydomonas: proportion of *C. reinhardtii* in the algal total abundance
- Relative_abundance_Monoraphidium_Chlorella: summed proportion of the species *M. minutum* and *C. vulgaris* in the algal total abundance
- "plankton_response_avgOEV_TC_share.csv" contains the algal and rotifer responses to the pulse perturbation estimated from the experimental timeseries, and the top-down control at the pulse, the relative biovolume of the different species at the pulse. There are 12 columns, among which the three first columns are Category, Treatment and Chemostat (cf. above) and the others are:
 - OEV_biovolume_algae: the algal biovolume response per day
 - OEV_biovolume_rotifers: the algal biovolume response per day
 - TC_at_pulse: the top-down control at the pulse
 - Relative_Biovolume_Monoraphidium_Chlorella_at_pulse: the summed share of *M. minutum* and *C. vulgaris* at the pulse
 - Relative_Biovolume_Chlamydomonas_at_pulse: the share of *C. reinhardtii* at the pulse

- Relative_Biovolume_Cryptomonas_at_pulse: the share of *Cryptomonas* sp. at the pulse
- Relative_Biovolume_Brachionus_at_pulse: the share of *B. calyciflorus* at the pulse
- Relative_Biovolume_Cephalodella_at_pulse: the share of *Cephalodella* sp. at the pulse
- Relative_Biovolume_Lecane_at_pulse: the share of *Lecane* sp. at the pulse
- "algae_no-rotifer_chemostat_preliminary_experiments.xlsx": contains the preliminary experimental timeseries of algal species and trophic level biovolumes and of the relative abundance of algal species. These experiments were conducted in absence of rotifers to have indicative insights about the algal species' behaviour. These datasets are not used in the main analyses of the study or are only used as supplementary material. The structure of the dataset is similar to the one described for "Chemostat_experimental_timeseries.xlsx".
- "algae_stoichiometry_preliminary_experiments.xlsx": contains the preliminary experimental timeseries of nitrogen content, carbon content, C/N ratio, cell volume, density, biovolume of algal species. These experiments were conducted in absence of rotifers to have indicative insights about the algal species' behaviour. These datasets are not used in the main analyses of the study or are only used as supplementary material. There are 9 columns:
 - Date: date of the measurement (DD/MM/YY)
 - Time standardised to pulse: day standardised to the day of the pulse perturbation. Negative values correspond to the days before the pulse perturbation, 0 corresponds to the day of the pulse perturbation and positive values correspond to the days after the pulse perturbation.
 - Algae: the algae species present in the experiment. "Cr" is *Cryptomonas* sp., "Ca" is *C. reinhardtii*, "Mo" is *M. minutum*, "Co" is *C. vulgaris* and "Mi" is the four algal species together.
 - N [pmol/cell]: nitrogen content in pmol/cell
 - C [pmol/cell]: carbon content in pmol/cell
 - C/N Ratio [molar]: ratio between N and C in [pmol/cell]/[pmol/cell]
 - Cell volume [μm^3]: algal cell volume in μm^3
 - Density [cells/mL]: algal density in cells/mL
 - Biovolume [$\mu\text{m}^3/\text{mL}$]: algal biovolume in $\mu\text{m}^3/\text{mL}$
- "fit_monocultures.rds" and "fit_polycultures.rds" are the files containing the a-priori distributions of the parameters of the equation-based model used for the Bayesian inference. They were used to generate the predicted timeseries of nitrogen dynamics, algal biovolume and rotifer biovolume of "Monocultures" and "Polycultures", that are contained in the files "bayesian_predictions_monocultures.csv" and "bayesian_predictions_polycultures.csv" , respectively.
- "bayesian_predictions_monocultures.csv": contains the predicted timeseries of nitrogen dynamics, algal biovolume and rotifer biovolume of "Monocultures". There are 11 columns:

- Chemostat: the name of the chemostat
- Time standardised to pulse: day standardised to the day of the pulse perturbation. Negative values correspond to the days before the pulse perturbation, 0 corresponds to the day of the pulse perturbation and positive values correspond to the days after the pulse perturbation.
- N.Q1 the first quantile value of the predicted nitrogen concentration in $\mu\text{mol N/L}$
- N.Q2 the second quantile value of the predicted nitrogen concentration in $\mu\text{mol N/L}$
- N.Q3 the third quantile value of the predicted nitrogen concentration in $\mu\text{mol N/L}$
- Biovolume.Q1_A the first quantile value of the predicted algal biovolume in $\mu\text{m}^3/\text{mL}$
- Biovolume.Q2_A the second quantile value of the predicted algal biovolume in $\mu\text{m}^3/\text{mL}$
- Biovolume.Q3_A the third quantile value of the predicted algal biovolume in $\mu\text{m}^3/\text{mL}$
- Biovolume.Q1_R the first quantile value of the predicted rotifer biovolume in $\mu\text{m}^3/\text{mL}$
- Biovolume.Q2_R the second quantile value of the predicted rotifer biovolume in $\mu\text{m}^3/\text{mL}$
- Biovolume.Q3_R the third quantile value of the predicted rotifer biovolume in $\mu\text{m}^3/\text{mL}$
- "bayesian_predictions_polycultures.csv": same as "bayesian_predictions_monocultures.csv" for "Polycultures".

Folder figures

This folder contains all the pdf files generated by the R scripts. In the following we provide the caption of the figure and which script was used to generate it.

- "figure2.pdf"
 - Caption: Relationships between a) $\log_2$ algal response (avgOEVA) or b) $\log_2$ rotifer response (avgOEVR) and top-down control TC. We fitted quadratic regressions for the 24 chemostats together (orange solid line), for the 12 rotifer monocultures (blue dotted curve and symbols: circles for *B. calyciflorus*, squares for *Cephalodella sp.* and diamonds for *Lecane sp.*), for the 12 rotifer polycultures (black dashed curve and symbols: down-triangles for chemostats perturbed 9 days after inoculation and up-triangles for chemostats perturbed 13 days after inoculation). The shadowed areas are the 90% credible intervals. Full summary statistics are available in the Appendix S5.
 - Script: "figure2.R"
- "figure3.pdf"
 - Caption: Timeseries of dissolved nitrogen concentration (dotted grey curve), algal biovolume (dashed green curve) and rotifer biovolume (solid magenta curve) in eight

chemostats: four rotifer monocultures of *Lecane* sp. (a), *Cephalodella* sp. (b), or *B. calyciflorus* (c and d), and four rotifer polycultures (e-h) differing in top-down control (TC) at the pulse ($t = 0$). Upper panels are chemostats with lower TC values. Dots are experimental data, whereas the curves and shaded areas show the respective means and 90% credible intervals of posterior predictive distributions from Bayesian inference. To fit the experimental timeseries, we first predicted pre-pulse dynamics and then post-pulse dynamics, which explains the jumps in the predictions at $t = 0$ (see details in the Appendix S4-S5). Timeseries of the other chemostats are available in the Appendix (Figs. S5.3-S5.4).

- Script: "figure3.R"
- "figure4.pdf"
 - Caption: (a) Per-capita net growth rates of rotifers predicted by Bayesian inference for *B. calyciflorus* monoculture (Bc) in dark red with circles, *Cephalodella* sp. monoculture (Ce) in red with squares, *Lecane* sp. monoculture (Le) in orange with diamonds, and rotifer polycultures in black (Poly), and (b) algal nutrient uptake rate in rotifer monocultures (Mono; solid, blue curve) and rotifer polycultures (Poly; dashed, black curve). Curves and shaded areas show means and 90% credible intervals, respectively, of posterior predictive distribution from Bayesian inference (see Appendix S4 for details).
 - Script: "figure4.R"
- "figureS2_1.pdf"
 - Caption: Biovolume timeseries in chemostats with monocultures of *C. vulgaris* (Co), *M. minutum* (Mo), **C. reinhardtii* *(Ca), and *Cryptomonas* sp. (Cr) and with all the algal species together (polyculture). In all the panels, solid and dashed curves distinguish two replicates and the vertical black line marks the day of the pulse. In polycultures *M. minutum* and *C. vulgaris* could not be distinguished (cf. Appendix S3), thus their biovolumes were summed (Mo+Co), and *Cryptomonas* sp. biovolume was not displayed because it was several orders of magnitude lower than the biovolume of the other species ($< 10^5 \mu\text{m}^3 \cdot \text{mL}^{-1}$), but this species persisted until the end of the experiment.
 - Script: "figureS2_1.R"
- "figureS2_2.pdf"
 - Caption: Timeseries of C:N ratios (a), cell densities (b), biovolumes (c) and cell volumes (d) for monocultures of *C. vulgaris* (Co in solid, light green curve), *M. minutum* (Mo in solid, dark green curve), *C. reinhardtii* (Ca in dashed, light blue curve) and *Cryptomonas* sp. (Cr in dashed, dark blue curve). Monocultures were cultivated semi-continuously and pulsed the day marked by a vertical, black line by increasing the nitrogen concentration from 80 to 400 $\mu\text{mol N/L}$.
 - Script: "figureS2_2.R"
- "figureS5_1.pdf"

- Caption: Biovolume timeseries of rotifer monocultures of *B. calyciflorus* (Bc1-4), *Cephalodella sp.* (Ce1-4) or *Lecane sp.* (Le1-4). Algae are in full dots and dashed curves – *M. minutum* and *C. vulgaris* in green, *C. reinhardtii* in light blue, *Cryptomonas sp.* in dark blue – and rotifers in empty dots and solid curves – *B. calyciflorus* in dark red, *Cephalodella sp.* in red and *Lecane sp.* in orange.
- Script: "figuresS5_1_and_S5_2.R"
- "figureS5_2.pdf"
 - Caption: Biovolume timeseries of rotifer polycultures differing in the timing of the pulse (9d or 13d after inoculation) and thus TC values (cf. Table S5.1). Colors and markers are the same as in Figure 3 and S5.1.
 - Script: "figuresS5_1_and_S5_2.R"
- "figureS5_3.pdf"
 - Caption: Biovolume timeseries of rotifer monocultures of *B. calyciflorus* (Bc1-4), *Cephalodella sp.* (Ce1-4) or *Lecane sp.* (Le1-4) fitted via Bayesian inference. Nitrogen concentration is in dotted grey curve, algal total biovolume is in dashed green curve and rotifer total biovolume is in solid magenta curve. The green empty circles and the magenta full circles are the experimental data of algae and rotifers, respectively.
 - Script: "figuresS5_3_and_S5_4.R"
- "figureS5_4.pdf"
 - Caption: Biovolume timeseries of rotifer polycultures differing in the timing of the pulse (9d or 13d after inoculation) fitted via Bayesian inference. The figure layout is the same as in Figures 3 and S5.3.
 - Script: "figuresS5_3_and_S5_4.R"
- "figureS5_5.pdf"
 - Caption: Differences in algal response (a) and rotifer response between rotifer monocultures (low diversity, in blue) and rotifer polycultures (high diversity, in black). No significant differences were observed (p-value > 0.05 for all the Wilcoxon sum rank tests).
 - Script: "figureS5_5.R"

Folder scripts

This folder contains all the R scripts for generating the figures and performing the analyses of this study. In the following we provide the aim of the script, and which datasets are uploaded and which figures or datasets are generated. The scripts are presented in the order that they should be run:

- "get_responses_to_pulse.R"

- Aim: For each chemostat experiment, calculate the biovolume and compositional responses of algae and rotifers to the pulse perturbation, the top-down control at the pulse perturbation and the relative biovolume of each species per trophic level.
 - Uploaded data: "Chemostat_experimental_timeseries.xlsx"
 - Generated output: "plankton_response_avgOEV_TC_share.csv"
- "get_bayesian_predictions_monocultures.R"
 - Aim: predicts the timeseries of the nitrogen concentration, algal total biovolume and rotifer total biovolume of each chemostat of the "Monocultures" treatment using Bayesian inference
 - Uploaded data: "fit_monocultures.rds"
 - Generated output: "bayesian_predictions_monocultures.csv"
- "get_bayesian_predictions_polycultures.R"
 - Aim: predicts the timeseries of the nitrogen concentration, algal total biovolume and rotifer total biovolume of each chemostat of the "Polycultures" treatment using Bayesian inference
 - Uploaded data: "fit_polycultures.rds"
 - Generated output: "bayesian_predictions_polycultures.csv"
- "figure2.R"
 - Aim: Investigating the relationships between plankton responses and the hypothesized drivers (top-down control, the relative biovolume of the opportunist species per trophic level)
 - Uploaded data: "plankton_response_avgOEV_TC_share.csv"
 - Generated output: "figure2.pdf"
- "figure3.R"
 - Aim: Plotting experimental timeseries and the associated predicted timeseries via Bayesian modelling for different chemostats
 - Uploaded data: "Chemostat_experimental_timeseries.xlsx", "bayesian_predictions_monocultures.csv", "bayesian_predictions_polycultures.csv"
 - Generated output: "figure3.pdf"
- "figure4.R"
 - Aim: plotting the nitrogen uptake rate of algae and the rotifer per-capital net growth rates predicted by Bayesian modelling
 - Uploaded data: "fit_monocultures.rds", "fit_polycultures.rds"
 - Generated output: "figure4.pdf"
- "figureS2_1.R"

- Aim: Plotting the biovolume timeseries of the preliminary experiments run in absence of rotifers
- Uploaded data: "algae_no-rotifer_chemostat_preliminary_experiments.xlsx"
- Generated output: "figureS2_1.pdf"
- "figureS2_2.R"
 - Aim: Plotting the timeseries of nitrogen content, carbon content, C/N ratio, cell volume, density, biovolume of algal species for the preliminary experiments run in absence of rotifers.
 - Uploaded data: "algae_stoichiometry_preliminary_experiments.xlsx"
 - Generated output: "figureS2_2.pdf"
- "figuresS5_1_and_S5_2.R"
 - Aim: Plotting the experimental biovolume timeseries of plankton species for all chemostats
 - Uploaded data: "Chemostat_experimental_timeseries.xlsx"
 - Generated output(s): "figureS5_1.pdf", "figureS5_2.pdf"
- "figuresS5_3_and_S5_4.R"
 - Aim: Plotting the fitted biovolume timeseries of plankton species for all chemostats
 - Uploaded data: "Chemostat_experimental_timeseries.xlsx", "bayesian_predictions_monocultures.csv", "bayesian_predictions_polycultures.csv"
 - Generated output(s): "figureS5_3.pdf", "figureS5_4.pdf"
- "figureS5_5.R"
 - Aim: Investigating the differences in plankton responses between rotifer monocultures and polycultures
 - Uploaded data: "plankton_response_avgOEV_TC_share.csv"
 - Generated output(s): "figureS5_5.pdf"